

ATM System – Business Analysis Document

Business Requirements, Process Analysis & Acceptance Criteria

1. Document Purpose

The purpose of this document is to capture and analyse the business requirements, functional scope, business rules, stakeholder expectations, risks and acceptance criteria for the ATM System. The document is written from a Business Analyst perspective and is intended to provide a common understanding between business stakeholders, product/project teams, development teams and implementation teams.

2. Business Objective

- Provide customers with secure and reliable access to essential banking services through an ATM.
- Enable customers to perform transactions such as cash withdrawal, balance inquiry and mini statement generation.
- Ensure secure authentication through card validation and PIN verification.
- Provide clear responses for successful, unsuccessful, cancelled and timed-out transactions.
- Support reliable communication between the ATM application and the backend banking system.
- Ensure the solution supports applicable banking, security, usability and accessibility expectations.

3. Stakeholders

Stakeholder	Interest / Responsibility
Customer	Uses ATM services and expects secure, accurate and understandable transactions.
Business / Product Owner	Defines business needs, priorities and expected customer outcomes.
Business Analyst	Elicits, analyses, documents and validates requirements and business rules.
Project Manager	Coordinates delivery, timelines, dependencies and approvals.
Development Team	Implements the agreed business and functional requirements.
UAT / Business Team	Validates that the solution meets business expectations.
Banking / Operations Team	Supports banking processes, transaction handling and operational readiness.

4. Scope

In Scope

- Card insertion and validation.
- PIN authentication and failed-attempt handling.
- Cash withdrawal.
- Balance inquiry.
- Mini statement.
- Funds transfer, where applicable.
- Cash/cheque deposit, where applicable.
- Receipt generation.
- Session timeout and transaction cancellation.
- Error handling and recovery.
- ATM application, hardware components and backend banking system integration.

- Security, performance, reliability, usability and accessibility considerations.

Out of Scope

- Core banking backend logic unless directly required for ATM integration.
- Physical manufacturing defects and ATM enclosure durability.
- Bank employee administrative functions.
- Customer service operations.

5. Business Process Overview

- 1. Customer Identification:** Customer inserts a valid card and the ATM validates the card.
- 2. Authentication:** Customer enters a PIN and the system authenticates the request.
- 3. Service Selection:** Customer selects the required banking service.
- 4. Transaction Validation:** The system validates account status, available balance and applicable transaction rules.
- 5. Transaction Processing:** The ATM communicates with the backend banking system where required.
- 6. Transaction Completion:** The requested service is completed and the customer receives an appropriate confirmation.
- 7. Receipt / Session Closure:** Receipt is generated where applicable and the session is securely ended or times out.

6. Functional Business Requirements

ID	Requirement
BR-01	The system shall validate an inserted card before allowing access to banking services.
BR-02	The system shall authenticate the customer using a valid PIN.
BR-03	The system shall support cash withdrawal subject to account balance, transaction limits and applicable business rules.
BR-04	The system shall provide balance inquiry information from the applicable banking source.
BR-05	The system shall support mini statement generation where the service is available.
BR-06	The system shall support funds transfer where enabled for the ATM and account type.
BR-07	The system shall support deposit functionality where enabled.
BR-08	The system shall generate a receipt where applicable and according to customer/service preferences.
BR-09	The system shall allow a customer to cancel an eligible transaction before completion.
BR-10	The system shall terminate an inactive session after the defined timeout period.
BR-11	The system shall provide understandable messages when a transaction cannot be completed.
BR-12	The system shall protect customer authentication and transaction information.

7. Business Rules

- Only valid and eligible cards should be allowed to proceed to authentication.
- Repeated unsuccessful PIN attempts may result in card retention or other defined security controls.
- Transactions must respect account balance, transaction limits and applicable banking rules.
- A transaction should not be presented to the customer as successful unless the required processing has completed successfully.
- A session should be securely terminated after customer cancellation, successful completion or inactivity timeout.

- Network failures and unavailable banking services should result in clear customer communication and appropriate transaction recovery handling.

8. Non-Functional Requirements

- **Security:** Authentication and customer data must be protected.
- **Performance:** Transactions should be processed within agreed business response times.
- **Reliability:** The ATM should handle service interruptions and recovery scenarios appropriately.
- **Usability:** Customer instructions and transaction messages should be clear and easy to understand.
- **Accessibility:** The solution should support applicable accessibility expectations.
- **Integration:** Communication with the backend banking system should be reliable and controlled.

9. Data and Business Scenarios

Scenario	Business Data / Condition
Valid transaction	Valid card, valid PIN and eligible account.
Invalid authentication	Invalid PIN or authentication failure.
Insufficient funds	Account balance is insufficient for requested withdrawal.
Card issue	Expired, blocked or otherwise ineligible card.
Security condition	Multiple failed PIN attempts requiring defined security handling.
Network interruption	ATM cannot communicate reliably with the banking system.
Transaction cancellation	Customer cancels an eligible transaction before completion.
Session timeout	Customer remains inactive beyond the agreed timeout period.

10. Integration & System Context

- ATM Application Software
- Card Reader
- Keypad
- Display Screen
- Cash Dispenser
- Receipt Printer
- Network Communication Module
- Backend Banking System Interface
- Core Banking Test Environment / Database where applicable

11. Acceptance Criteria

1. Customer with a valid card and PIN can access eligible ATM services.
2. Invalid authentication is handled securely and the customer receives an understandable message.
3. Cash withdrawal is completed only when the applicable business conditions are satisfied.
4. Balance inquiry and mini statement provide the expected account information when the services are available.
5. The customer can cancel an eligible transaction and the session closes appropriately.
6. Inactive sessions are terminated after the agreed timeout period.

7. Expired or blocked cards are handled according to defined business rules.
8. Multiple failed PIN attempts trigger the defined card/security handling.
9. Network failure scenarios do not result in an incorrect customer-facing success outcome.
10. Receipt generation occurs according to the agreed transaction and business rules.

12. Risks and Mitigation Considerations

Risk	Business Impact	Mitigation Consideration
Network failure	Transactions may be interrupted or delayed.	Define recovery, timeout and customer communication rules.
Security breach	Customer data and account security may be compromised.	Apply authentication, security controls and appropriate security assessment.
Hardware malfunction	Customers may be unable to complete transactions.	Define monitoring, fallback and operational recovery procedures.
Cash dispensing error	Customer balance and cash reconciliation may be affected.	Define transaction reconciliation and exception-handling processes.

13. Assumptions and Dependencies

- The ATM is connected to the required banking network.
- Test accounts and transaction data represent the expected business behavior.
- ATM hardware components operate according to their specified capabilities.
- Applicable banking and security policies will be provided or confirmed by the relevant business stakeholders.
- Transaction limits, timeout values and card-handling rules will be confirmed as part of detailed requirements.

14. Business Analyst Deliverables

- Business Requirements Document (BRD) / Business Requirements.
- Functional requirements and business rules.
- As-is / To-be process flows where required.
- Use cases and user stories.
- Acceptance criteria.
- Requirements traceability and stakeholder clarification records.
- UAT scenarios and business validation support.

15. Conclusion

The ATM System should provide customers with secure, reliable and understandable access to banking services. From a Business Analyst perspective, the key focus is to establish clear business requirements, define transaction rules and expected outcomes, identify dependencies and risks, and ensure that the delivered solution supports the intended customer and business processes. Detailed technical testing activities can then be derived from these agreed requirements.